

MDWD: A Street-Level Dataset for Municipal Solid Waste Detection in Dense Urban Environments



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INTRODUCTION

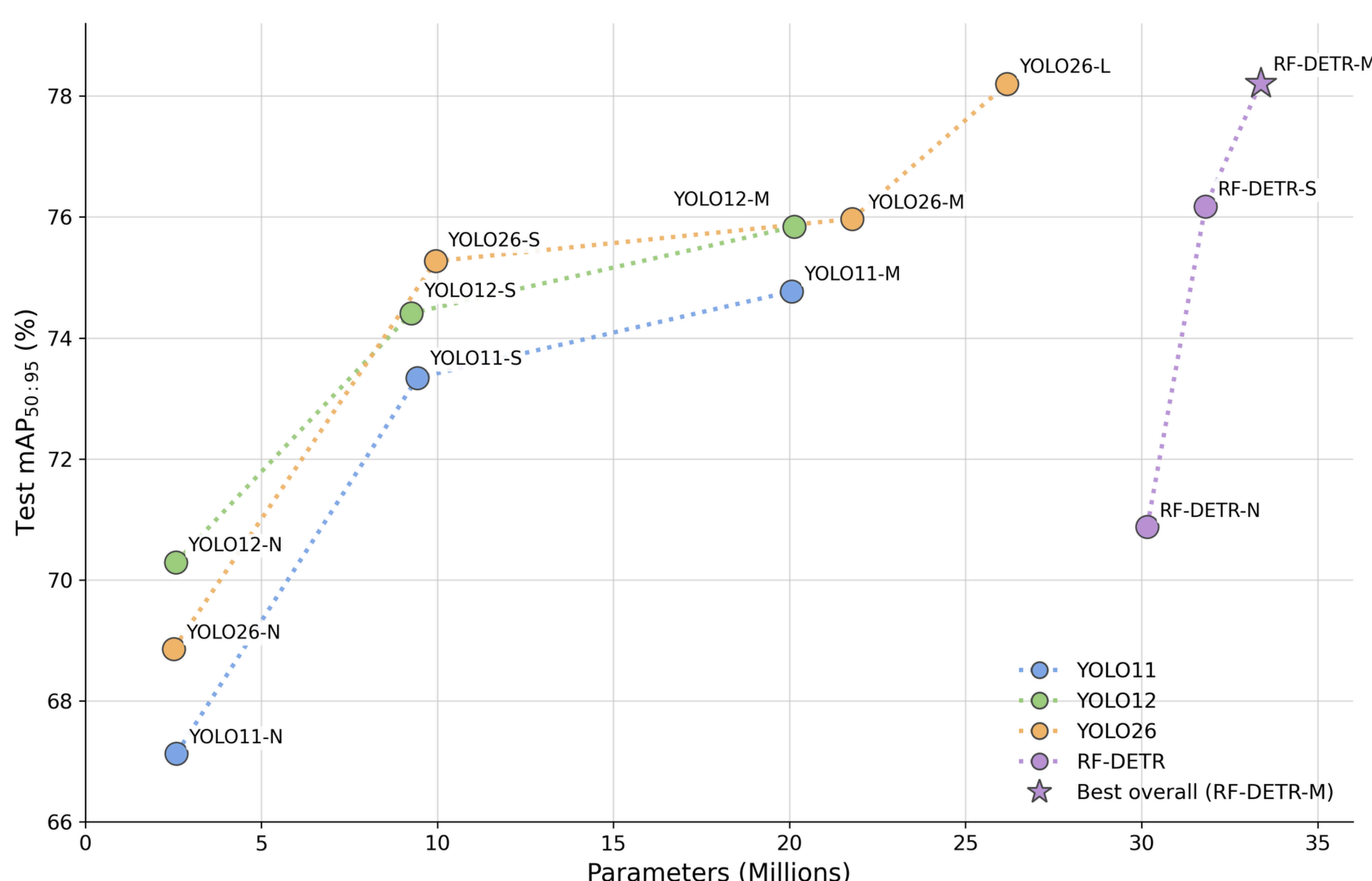
Municipal waste monitoring remains largely dependent on manual inspection and citizen reporting. Existing waste datasets predominantly address litter, aerial imagery, or image-level classification, leaving a gap for operational street-level monitoring.

- The **Maltese Domestic Waste Dataset (MDWD)** addresses this gap through street-level, instance-level **detection of municipal solid waste (MSW)** streams.
- It is designed to reflect realistic challenges, including occlusion, scale variation, illumination changes, and urban clutter.
- The work was funded by the **Government of Malta's Cleansing and Maintenance Division (CMD)**.

DATASET CONSTRUCTION & ANNOTATION

- MDWD comprises **3,697** high-resolution images collected across **Malta and Gozo** between winter 2024 and late summer 2025, with **11,461** annotated instances across the five waste categories.
- Instances were annotated using axis-aligned **bounding boxes** fitted to the visible object extent.
- Visible personal data, including identifiable faces and vehicle registration plates, were **redacted in accordance with GDPR** requirements.
- Annotations were produced by a trained team using **Roboflow** and subsequently reviewed by an experienced lead annotator for quality assurance.
- The dataset was split **80/10/10** into **2,958** training, **370** validation, and **369** test images.
- **Offline augmentations** were applied only to the training partition, expanding to **29,580** images using geometric, photometric, blur, and 2×2 mosaic transformations.

CROSS-ARCHITECTURE RESULTS



Parameter count versus test $mAP_{50:95}$ across evaluated detectors.

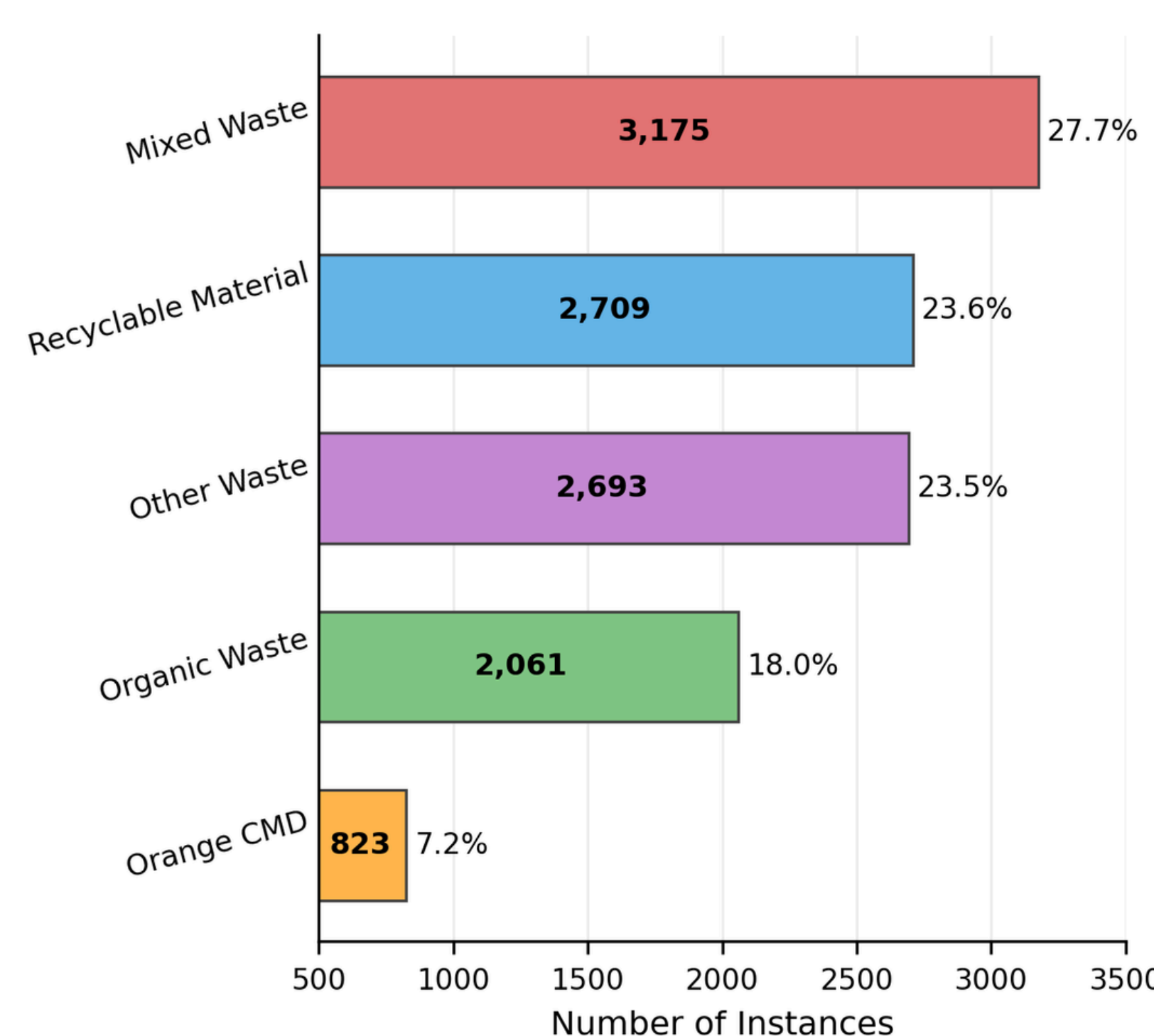
RF-DETR-M achieved the **strongest overall test-set performance**, attaining 94.49% mAP_{50} , 78.21% $mAP_{50:95}$, 96.63% precision, 90.69% recall, and 93.56% F1-score, while the **YOLO** variants **remained competitive** at substantially lower parameter counts.

THE MALTESE DOMESTIC WASTE DATASET

MDWD defines five visually and operationally distinct waste categories aligned with Malta's municipal collection system:

- **Mixed Waste** – Residual household refuse, predominantly presented in black bags.
- **Recyclable Material** – Recyclable packaging and materials, typically presented in gray or green bags.
- **Organic Waste** – Biodegradable household waste, typically presented in white bags under the national organic collection scheme.
- **Orange CMD** – Waste collected during municipal cleansing rounds by the CMD and presented in distinctive orange bags.
- **Other Waste** – Infrequently occurring or heterogeneous waste, including bulky items and loose litter such as glass bottles.

The taxonomy reflects operational collection streams rather than simply the material or object type, making the dataset directly relevant to operational municipal monitoring.

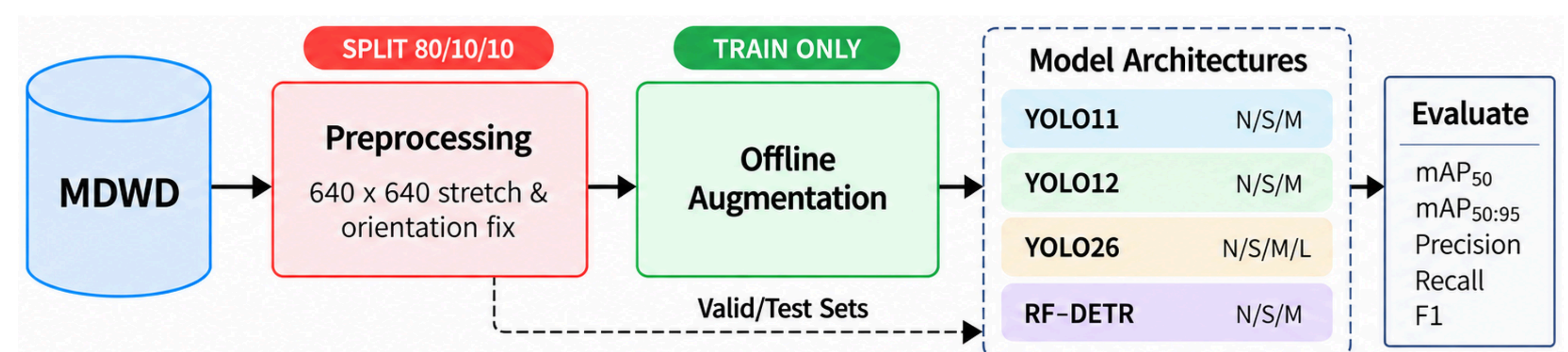


Distribution of annotated instances across the five classes.



Representative cropped MDWD annotation sample.

BENCHMARK METHODOLOGY



Overview of the benchmark methodology and evaluation pipeline.

CONCLUSION & FUTURE WORK

MDWD establishes a **reproducible street-level benchmark** for multiclass MSW detection across **convolutional and transformer-based** detectors. Lower performance on **Orange CMD** and especially **Other Waste** highlights the challenge of underrepresented and heterogeneous classes, while close validation–test agreement supports **generalization to unseen imagery**. Future work may include:

- Cross-region generalization and finer-grained waste taxonomies.
- Exploring promptable zero-shot localization and self-supervised visual representations.
- Developing saliency-based prioritization for operational municipal monitoring.

MDWD is publicly available on Roboflow:

